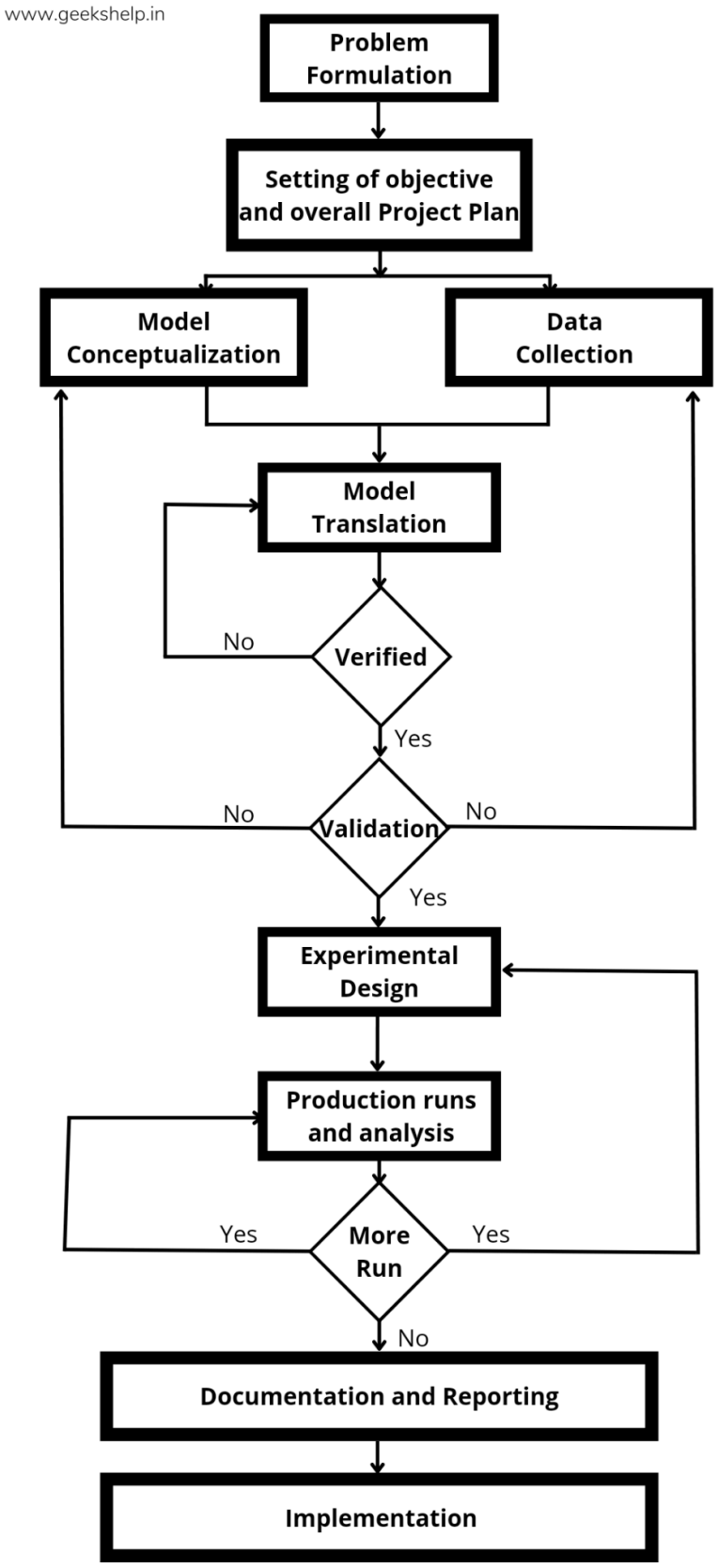
Steps in a simulation Study



1. Define the Problem: Start by clearly stating what problem you're trying to solve. If someone else gives you the problem, make sure you understand it. If you're figuring it out yourself, make sure others agree with your understanding.

2. Set Goals and Make a Plan: Figure out what you want to learn from the simulation. Plan out how you'll do it, including what different options you'll consider, how you'll measure success, and how long it will take.

3. Build the Model: Making a model is like making a simplified version of the real problem. It's both an art and a science. You start simple and add detail as needed. It's also important to involve others who will use the model.

4. Collect Data: As you build your model, you'll need information to make it accurate. You have to determine what type of data to collect. Start gathering data early, and make sure it matches what you need to know. Data is fitted to theoretical distributions. For example, the arrival rate of a specific part to the manufacturing plant may follow a normal distribution curve. The collection of these distributions is known as an Input Model.

5. Translate the model: Models can get complicated, so you need to make them computer-friendly. This step involves putting your model into a format that a computer can understand. You might need to do some programming.

6. Verified: This step ensures that the computer program used for the simulation model is working correctly. It involves checking if the program accurately represents the model's input parameters and logical structure. Sometimes, debugging is needed to fix any issues.

7. Validated: Validation is done by comparing the simulation model against real-world system behaviour. This comparison helps to improve the model's

accuracy through iterative adjustments until it closely matches reality. For example, in a bank simulation, validation might involve comparing simulated waiting line lengths with actual waiting times.

8. Experimental Design: This step involves determining which alternatives to simulate. It considers factors like the length of simulation runs, the number of replications, and the initialization period.

9. Production Runs and Analysis: This phase runs simulations and analyses the results to estimate system performance measures. Special software like AutoStat, OptQuest, and SimRunner can help with this analysis.

10. More Runs: Based on the analysis of completed runs, the analyst decides if additional simulations are needed and plans their design.

11. Documentation and Reporting: This involves documenting the simulation program and progress made in the project. It helps in understanding and modifying the program, keeping track of project milestones, and reporting results clearly in a final report.

12. Implementation: The success of this phase depends on the effectiveness of the previous steps. It involves putting the simulation model into action, and effective communication with the model user is crucial for successful implementation.